

Fig. 9: Close-up of the frequency counter PCB



Fig. 10: The chip-on-board is the UTC6610

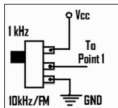


Fig. 11: Connect a small switch between point '1' and GND and Vcc

(HF operation). Setting it to Vcc in HF mode only sets the display resolution to 1 kHz. '2' is an output point for the radio which lets the IC know whether it's the AM (SW/MW) mode or FM mode. '3' is an input point at Vcc (3V, do not exceed 3.3V). '4' is the input point at GND. '5' is the output point that controls power to the radio, shutting off power to the IC when in clock mode.

Pressing FM button activates the VHF counter with a resolution of 100 kHz and frequency range of 11 MHz to 150 MHz. The input is via the purple wire (shown in Fig. 9); the colour may differ in your radio. You must set point 1 to 0V for proper operation. The display reading must be added to 10.7 MHz to get the actual frequency.

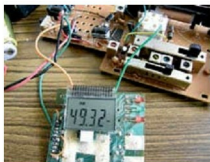


Fig. 12: Crude connection example

A flip-flop comprising SMD transistors on the PCB locks the UTC in this mode until the AM button is pressed. Default mode on startup is the FM mode.

Pressing AM button activates the HF counter with frequency range of 0.5 to 30 MHz. The input is via the black wire (shown in Fig. 9); the colour may differ in your radio. Set point 1 to Vcc for 1kHz resolution and RF input of 0.5 to 9.999 MHz. Set point 1 to 0V for 10kHz resolution and 0.5 MHz to 29.99 MHz. Actual tests show it exceeds 75 MHz in this mode, which is good. Reading on the display must be added to 455 kHz to get the actual frequency.

A flip-flop comprising SMD transistors on the PCB locks the UTC in this mode until the FM button is pressed. Default mode on startup is the FM mode. To activate the AM mode, this button has to be pressed.

Notice the blob of solder above the VHF input point. This is the ground.

Of course, the clock pushbuttons are still working perfectly, so you may use them to set the clock too. You will have to shift the SPDT switch (the switch which controls the radio 'on' and 'off') to 'off' position in order to activate the clock, and to 'on' position in order to activate the RF counter.

Please note that here we're not trying to use the UTC6610 directly. Most of the hard work of wiring up the 6610 IC, providing an LCD, making the PCB and adding components has been done by the radio manufacturer; in this case, Kchibo. All we are attempting to do here is to understand operation of the UTC6610, and in turn operation of the

PCB provided by the manufacturer of the radio, so as to come up with a quick, compact, cost-effective RF counter.

To have a look at the IC, lift the LCD, and you will see that the chip-on-board (COB) is the UTC 6610 (see Fig. 10).

Making the RF counter

1. Cut out the 5-wire ribbon cable. Unsolder it from the counter board.

2. Cut the VHF and HF input wires from the points on the radio PCB, not the counter PCB.

3. The counter PCB is now separated from the radio receiver PCB.

4. Solder a red wire to point '3' of the counter PCB. This is your positive 3V supply wire.

5. Solder a green wire to point '4' of the PCB. This serves as ground.

6. Connect a small switch between point '1' and GND and Vcc, as shown in Fig. 11. Push the switch to 1kHz position for 1kHz resolution in the AM mode, and to 10kHz resolution for 10kHz in the AM mode. The switch has to be kept in the 10kHz resolution mode for proper VHF operation (FM mode).

With this, your crude-looking but great-performance RF counter is set to go. But bear a few things in mind:

1. Keep the VHF and HF wires as short as possible. Their length will not affect the performance of the counter, but the circuit under test (for example, the oscillator you might have built).

2. Couple the HF and VHF wires to the RF sources via a 2pF capacitor.

3. Preferably, replace the VHF and HF crude wires with easily available 75-ohm cable TV coax. This provides RF shielding and noise immunity to a good degree.

Fig. 12 shows a crude connection example. Note how the green wire (HF wire) is directly fed to the output of the oscillator (very crude, do not attempt this), without any capacitor or buffer. (I admit this green wire belongs to a different counter, but the operation and concept are the same, as it too uses the UTC6610.) Once I disconnected the counter from the oscillator I had made,